The Omics Hub: Bioinformatics & Command Line Cheat Sheet

The Omics Hub

2026

Command Line Basics

Navigation

- ‘pwd‘: Print working directory (where am I?)
- ‘ls -lh‘: List files with human-readable sizes
- ‘cd /path/to/dir‘: Change directory
- ‘cd ..‘: Move up one directory
- ‘cd ~‘: Move to home directory

File Operations

- ‘cp file1 file2‘: Copy file1 to file2
- ‘mv old_name new_name‘: Move or rename a file
- ‘rm file.txt‘: Remove/delete a file (CAUTION: irreversible)
- ‘mkdir my_analysis‘: Create a new directory
- ‘rm -r my_analysis‘: Remove a directory and its contents

Examining Data

- ‘head -n 10 data.fastq‘: View first 10 lines
- ‘tail -n 10 data.fastq‘: View last 10 lines
- ‘less data.fastq‘: View a file page-by-page (press ‘q‘ to exit)
- ‘wc -l file.txt‘: Count the number of lines in a file

Searching & Filtering (The Bioinformatics Swiss Army Knife)

- ‘grep “pattern” file.txt‘: Find lines containing “pattern”
- ‘grep -c “pattern” file.txt‘: Count occurrences of “pattern”
- ‘grep -v “pattern” file.txt‘: Find lines that DO NOT contain “pattern”
- ‘awk ‘{print $1, $3}’ file.txt‘: Print 1st and 3rd columns
- ‘sort -k 2,2n file.bed‘: Sort a BED file numerically by the second column

High-Performance Computing (HPC) / Slurm

Job Management

- ‘squeue -u username‘: Check status of your jobs

- ‘scancel 123456‘: Cancel job ID 123456
- ‘sbatch my_script.sh‘: Submit a batch job

Example Slurm Script Header

“‘bash #!/bin/bash #SBATCH –job-name=my_analysis #SBATCH –nodes=1 #SBATCH –cpus-per-task=8 #SBATCH –mem=32G #SBATCH –time=24:00:00 #SBATCH –output=my_analysis_%j.log “‘

Conda / Mamba Environment Management

- ‘conda create -n rna_seq_env python=3.9‘: Create a new environment
- ‘conda activate rna_seq_env‘: Activate environment
- ‘conda install -c bioconda fastqc‘: Install a tool from bioconda channel
- ‘conda env list‘: List all environments
- ‘conda list‘: List installed packages in current environment

Key Bioinformatics File Formats

1. **FASTA (.fa, .fasta)**: Reference genomes, transcriptomes. Alternating headers (‘>seq1‘) and sequences.
2. **FASTQ (.fq, .fastq)**: Raw sequencing reads with quality scores.
3. **BAM/SAM (.bam, .sam)**: Aligned reads. SAM is plain text; BAM is compressed binary.
4. **VCF (.vcf)**: Variant Call Format. Contains SNPs and Indels.
5. **GTF/GFF (.gtf, .gff)**: Gene annotations (coordinates of exons, genes, CDS).

Quick Tips

- **Piping (‘|‘)**: Pass the output of one command to another. Example: ‘ls -l | wc -l‘
- **Redirection (‘>‘)**: Save output to a file. Example: ‘echo “Hello” > file.txt‘
- **Appending (‘»‘)**: Add output to the end of a file. Example: ‘echo “World” » file.txt‘
- Use ‘tab‘ for auto-completion to avoid typos!

Important Single-Cell / Biomarker Genes

Below is a quick reference for common biomarkers used in immunology and cancer research, specifically relating to T-cell malignancies like Sézary Syndrome:

- **CD4**: Helper T-cell marker.
- **FOXP3**: Master regulator for Regulatory T cells (Tregs). In Sézary syndrome, differentiating CD25+ and CD25- phenotypes is critical.
- **CLIC1**: Chloride Intracellular Channel 1. A key marker associated with metabolic vulnerabilities in malignant cells.
- **CD25 (IL2RA)**: Alpha chain of the IL-2 receptor, typically expressed on Tregs and activated T cells.
- **CCR4 / CCR7**: Chemokine receptors important for T-cell skin homing and lymph node migration.